Huawei Access Node HEB-ACC-33 Hebbal Edge Optical Transport Failure ALARM_8454 Minor Severity Standard Operating Procedure (SOP)

Node: HEB-ACC-33 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_8454

Description

The Optical Transport Failure in the Huawei Access Node HEB-ACC-33 at the Hebbal Edge site is attributed to the degradation of the SFP/XFP interface or a fiber cut, leading to a decrease in Rx/Tx power loss. This failure domain occurs in the Access role due to the reliance on Optical Transport Network (OTN) protocols such as G.709 and G.783 for the transport of client signals over the Optical Transport Network (OTN) infrastructure. The SFP/XFP interface, specifically designed for high-speed data transmission, can be susceptible to degradation due to factors like aging, contamination, or physical damage. Furthermore, the OTN protocols employed by Huawei's equipment, such as the NE80E and NE40E series, are prone to errors in cases of fiber cuts or Rx/Tx power loss. The ALARM_8454, indicating a minor severity, signifies that the failure is not catastrophic, but it necessitates prompt attention to prevent further degradation and potential service disruptions.

Pre-requisite Checks

- Pre-requisite: Verify the redundant link status and ensure it is operational.

- Pre-requisite: Gather the necessary equipment and personnel to perform the diagnostic and resolution steps.

Potential Root Causes

Cause	Probability
Degradation of SFP/XFP interface.	High
Fiber cut.	Medium
Rx/Tx power loss.	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical-otr alam, Expected Output: ALARM_8454: Minor, Optical Transport Failure., Description: This command displays the alarm information for the Optical Transport Failure, including the alarm code, severity, and location.

Command: display optical-otr trr, Expected Output: Rx/Tx power loss: 0.5 dB., Description: This command displays the Tx/Rx power loss information for the Optical Transport Network, helping to identify potential issues with the fiber link.
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Resolution Steps

Step 1: Step: Inspect the SFP/XFP interface., Command: display optical-otr sfp, Description: This step involves visually inspecting the SFP/XFP interface for any signs of degradation, contamination, or physical damage.

Step 2: Step: Replace the SFP/XFP interface., Command: replace optical-otr sfp , Description: If the SFP/XFP interface is found to be degraded or damaged, this step involves replacing it with a new one to restore normal operation.

Step 3: Step: Verify the fiber link., Command: display optical-otr trr, Description: This step involves verifying the fiber link to ensure there are no issues with the Rx/Tx power loss.

Post-Check Verification

```
Command: display optical-otr alam, Expected Output: ALARM_8454: Minor, Optical Transport Failure., Description: This command verifies that the alarm has cleared after the resolution steps.

Command: display optical-otr trr, Expected Output: Rx/Tx power loss: 0 dB., Description: This command verifies that the Tx/Rx power loss has returned to normal after the resolution steps.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP interface., Command: restore optical-otr sfp , Description: If the fix fails, this step involves restoring the original SFP/XFP interface to its previous state.

- Step: Re-inspect the fiber link., Command: display optical-otr trr, Description: This step involves re-inspecting the fiber link to ensure there are no issues with the Rx/Tx power loss.